Language Exclusion Risk Score — Methodology

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Status: Formula finalized. Awaiting cleaned dataset (Shammah) for final calculation and validation.
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1. Purpose

The Language Exclusion Risk Score measures how likely a country's population is to be excluded from digital identity systems and e-government services because those systems operate only in the official/colonial language, not in the languages people actually speak day to day.

A higher score = higher risk of exclusion. This score is the core analytical output of the Language Mapping workstream and feeds directly into the dashboard's map and ranking visuals.

2. Required Input Columns

These columns must exist in the cleaned dataset before this formula can be calculated. Handed off by Shammah / Data Strategy team:

Column	Description	Expected format
Country	ECOWAS member state name	Text, standardized spelling
Adult Literacy %	National adult literacy rate	Decimal number, 0–100
Official Language Fluency %	% of population fluent in the official language	Decimal number, 0–100 (derived by Naimat from

Column	Description	Expected format
		Notes column / secondary sources)
Internet Users %	% of population using the internet	Decimal number, 0–100
Indigenous Language Count	Number of widely-spoken indigenous languages with no official digital recognition	Whole number

If any of these are missing or still in "value (year)" text format (e.g., "41.2 (2024)") when handed off, they must be split and converted to Decimal Number in Power Query before this formula can be applied.

3. Formula

```
Language Exclusion Risk Score = Literacy Risk + Fluency Risk +
Internet Risk + Fragmentation Risk
```

Each component is scored on a 0–25 scale, so the total score ranges from 0 (no risk) to 100 (maximum risk).

3.1 Literacy Risk (0–25)

```
Literacy Risk = 25 × (1 − Adult Literacy % / 100)
```

Lower literacy → higher risk. A country with 100% literacy scores 0 here; a country with 0% literacy scores the full 25.

3.2 Fluency Risk (0–25)

```
Fluency Risk = 25 × (1 − Official Language Fluency % / 100)
```

Lower fluency in the official language (the language digital ID systems actually use) → higher risk.

3.3 Internet Risk (0–25)

```
Internet Risk = 25 × (1 − Internet Users % / 100)
```

Lower internet access → higher risk, since digital ID systems assume connectivity.

3.4 Fragmentation Risk (0–25)

```
Fragmentation Risk = 25 × MIN(Indigenous Language Count / 10, 1)
```

More indigenous languages with no official digital recognition → higher fragmentation risk. Capped at 10 languages so countries with extremely high language counts (e.g., Nigeria with 500+) don't distort the scale; anything at or above 10 receives the maximum 25.

4. Risk Tiers

Score Range	Tier
0 – 33	Low
34 – 66	Medium
67 – 100	High

5. Why These Four Indicators (Rationale)

- **Literacy and Fluency are kept separate** because a person can be fully literate in their mother tongue while still being excluded from a digital ID system that only operates in the official language. Combining them into one indicator would hide this distinction, which is the core finding of this project.
- **Internet access** is included because even a perfectly translated digital ID system is useless to someone who cannot get online — this reflects the "digital" half of "digital identity."
- **Fragmentation** captures a risk that literacy/fluency/internet alone miss: a country can have decent official-language fluency overall while still having millions of speakers of unrecognized indigenous languages who are invisible in national averages.

- **Equal weighting (25 each)** was chosen for simplicity and defensibility at this stage. If time allows, a sensitivity analysis (re-running the score with different weights) can be added to show the ranking is robust — see Section 7.

6. Illustrative Example (Using Preliminary, Pre-Cleaning Estimates)

These figures are from earlier working data and are **illustrative only** — they must be recalculated once Shammah's cleaned dataset is finalized.

Country	Literacy Risk	Fluency Risk	Internet Risk	Fragmentation Risk	Total Score	Tier
Nigeria	5.4	~19.0 (est. 24% fluency)	14.7	25 (7+ major languages)	~64.1	Medium
Ghana	4.7	~14.0 (est. 44% fluency)	7.0	20 (8 languages)	~45.7	Medium
Senegal	9.6	~19.5 (est. 22% fluency)	10.0	15 (6 languages)	~54.1	Medium

Sanity check to re-run once real data lands: Senegal and Guinea are expected to land in the **High** tier given very low official-language fluency combined with lower literacy — if the final numbers don't reflect this, double-check the formula direction before finalizing.

7. Validation Plan

- [ ] Cross-check Official Language Fluency % extractions against a second source (Ethnologue, national census, or CLEAR Global) for at least 3 countries

- [ ] Confirm no column is still in text format before calculating (check for a green number icon in Power BI's column header)
- [ ] Re-run the sanity check in Section 6 once real cleaned data is in
- [ ] Optional: test with weights of 30/30/20/20 or similar to confirm country rankings don't dramatically shift — include this as a footnote in the final presentation if time allows, to preempt judge questions about weighting choices

8. Handoff to Blessing (DAX & Calculations)

Once the cleaned dataset is loaded, the following DAX measures should be created exactly as follows (column names in brackets should be adjusted to match the final cleaned table's exact column names):

```
Literacy Risk = 25 * (1 - ('ECOWAS Master Data'[Adult Literacy
%] / 100))

Fluency Risk = 25 * (1 - ('ECOWAS Master Data'[Official Language
Fluency %] / 100))

Internet Risk = 25 * (1 - ('ECOWAS Master Data'[Internet Users
%] / 100))

Fragmentation Risk = 25 * MIN('ECOWAS Master Data'[Indigenous
Language Count] / 10, 1)

Language Exclusion Risk Score = [Literacy Risk] + [Fluency Risk]
+ [Internet Risk] + [Fragmentation Risk]

Risk Tier = SWITCH(
    TRUE(),
    [Language Exclusion Risk Score] <= 33, "Low",
    [Language Exclusion Risk Score] <= 66, "Medium",
    "High"
)
```

Please flag me (Naimat) if any input column doesn't exist yet or is named differently — the formula depends on the four input columns in Section 2 being clean, numeric, and correctly named.

9. Data Sources

- Ethnologue (language mapping, speaker estimates)
- UNESCO World Atlas of Languages / UNESCO Institute for Statistics
- World Bank Open Data (literacy, internet access indicators)
- Team-compiled Official Language Fluency estimates (see Section 2 note; cross-validated against at least one secondary source per country)